Transcriptomic Profiling of Vitamin D–Related Compounds

Exploratory analysis of L1000 transcriptomic signatures across compounds, cell lines, and latent clusters.

- **258** transcriptomic signatures
- **12** vitamin D–related compounds
- **5** human cell lines
- **5** transcriptomic clusters (unsupervised)

Overview | Quality Control | Transcriptomic Structure | Clusters & Compounds

Data Science & Bioinformatics Portfolio
Paula Irazoqui

Overview — Dataset Scope

Filters

Compounds Name

Todas

Cell Lines

Todas

Dose

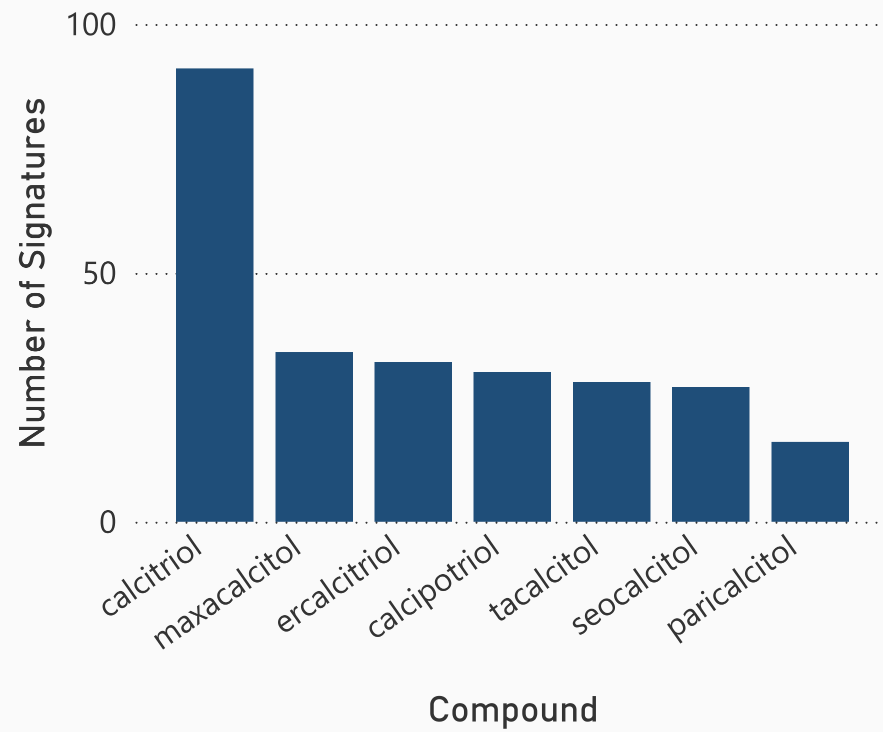
0,01 | 10,00

Clusters

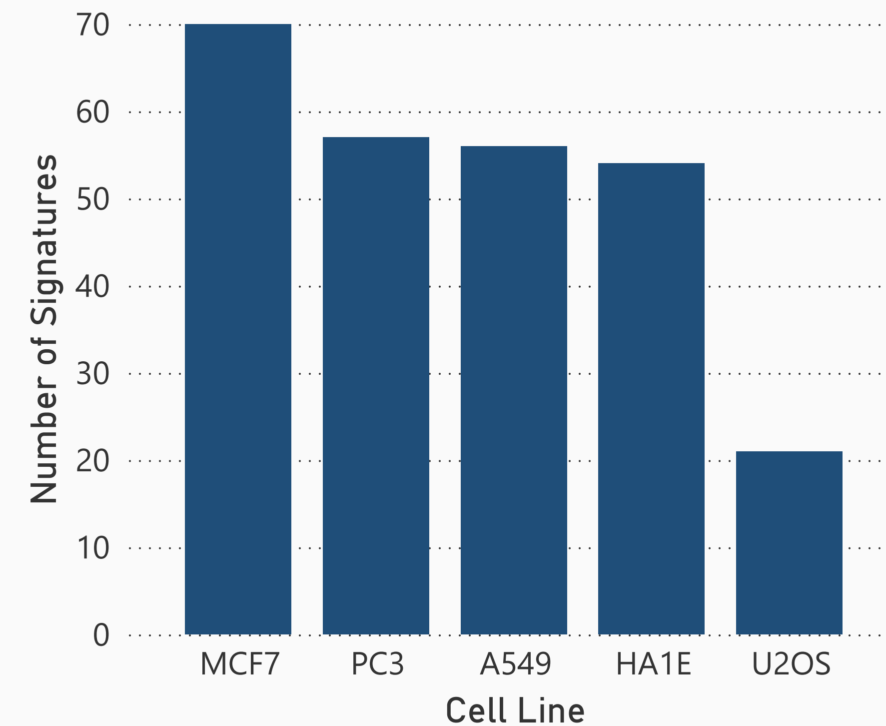
0 | 4

Signatures Analyzed	Vitamin D–related Compounds	Transcriptomic Clusters	Cell Lines
258	12	5	5

Distribution of Signatures per Compound



Signature Distribution across Cell Lines



Quality Control — Signature-Level Metrics

Filters

Compounds Name

Todas

Cell Lines

Todas

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0,01 | 10,00

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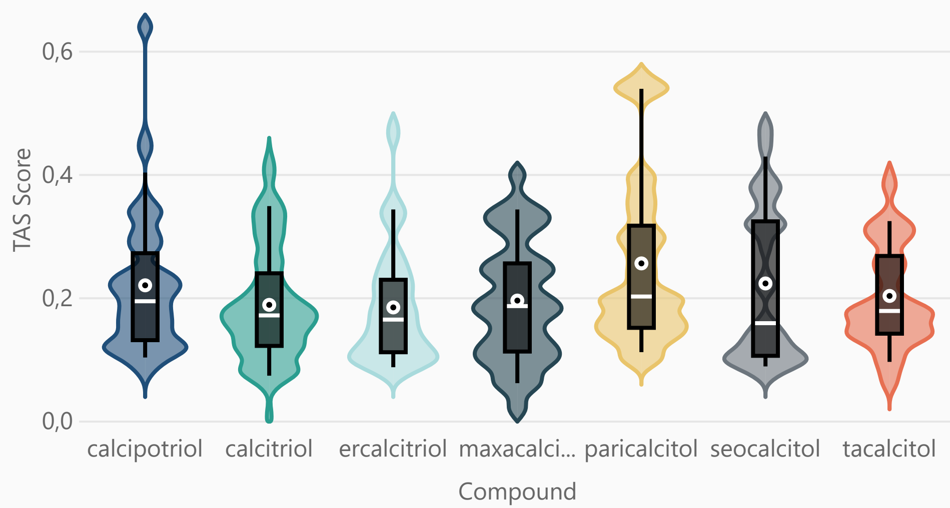
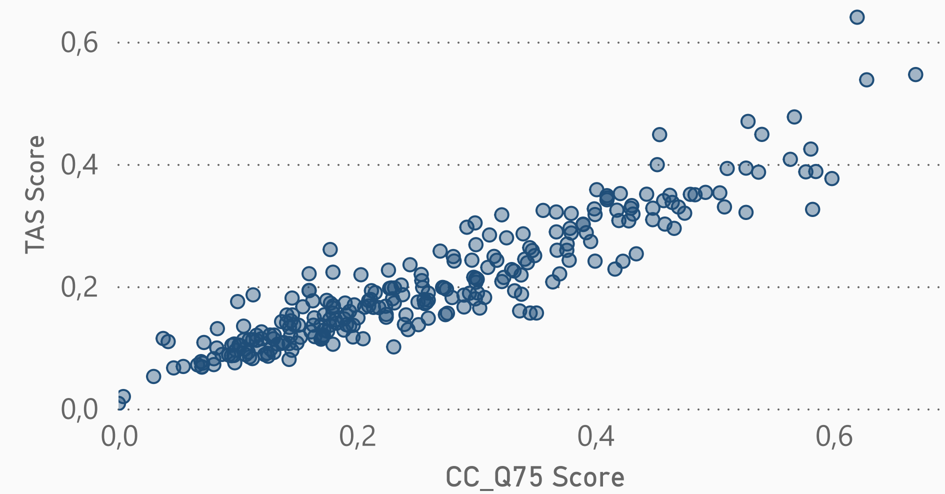
0 | 4

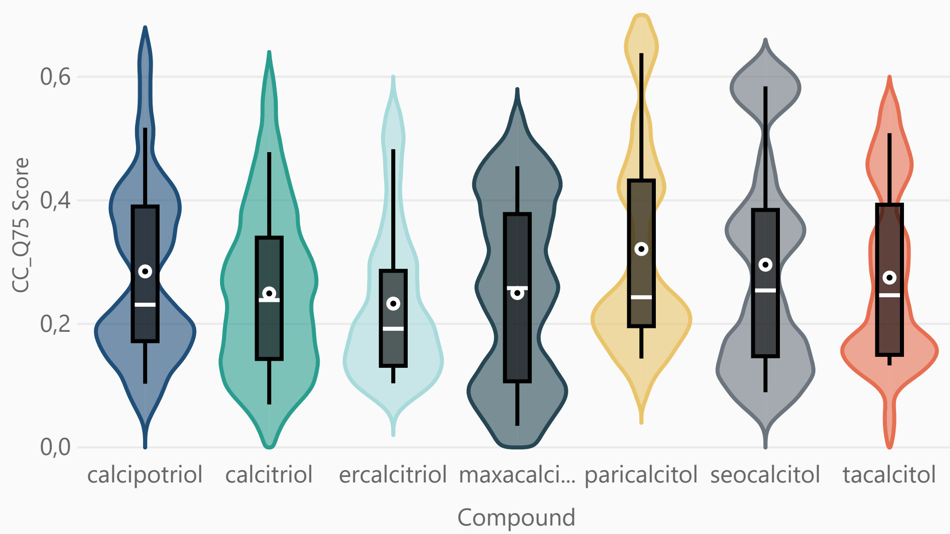
TAS Distribution per Compound



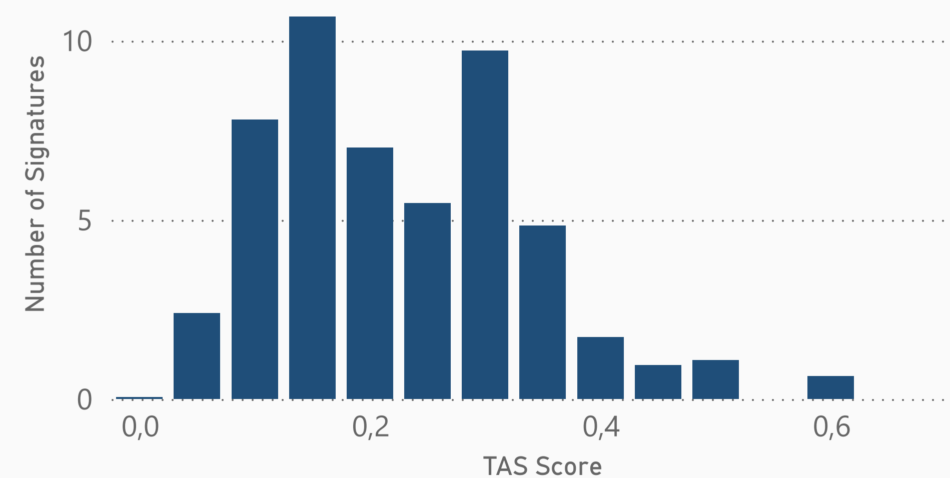
CC_Q75 vs TAS Correlation



CC_Q75 Distribution per Compound



TAS Score Distribution



Transcriptomic Structure — Latent Organization of Signatures

Filters

Compounds Name

Todas

Cell Lines

Todas

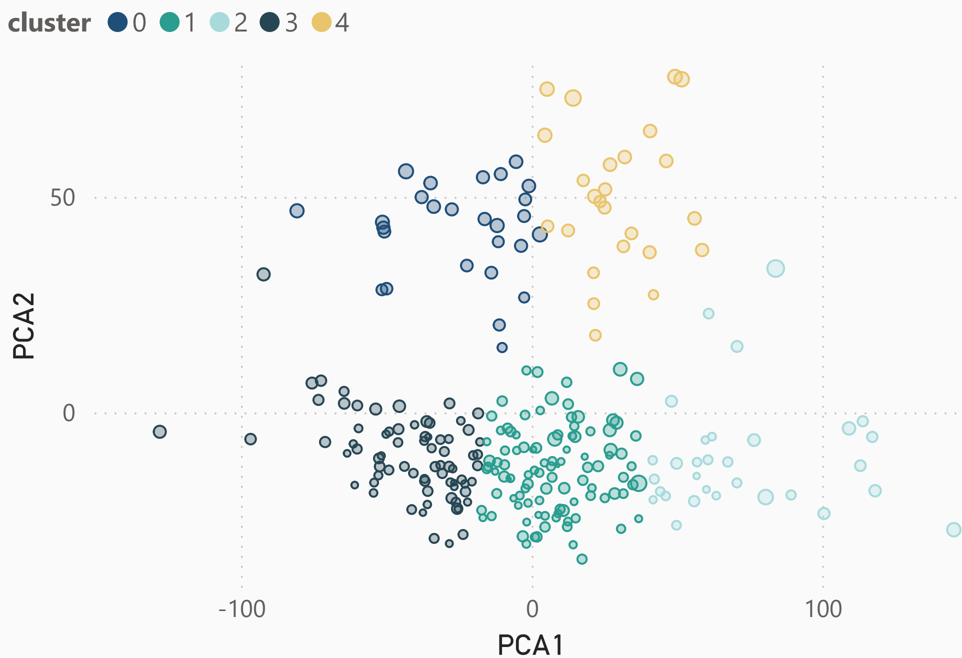
Dose

0,01 | 10,00

Clusters

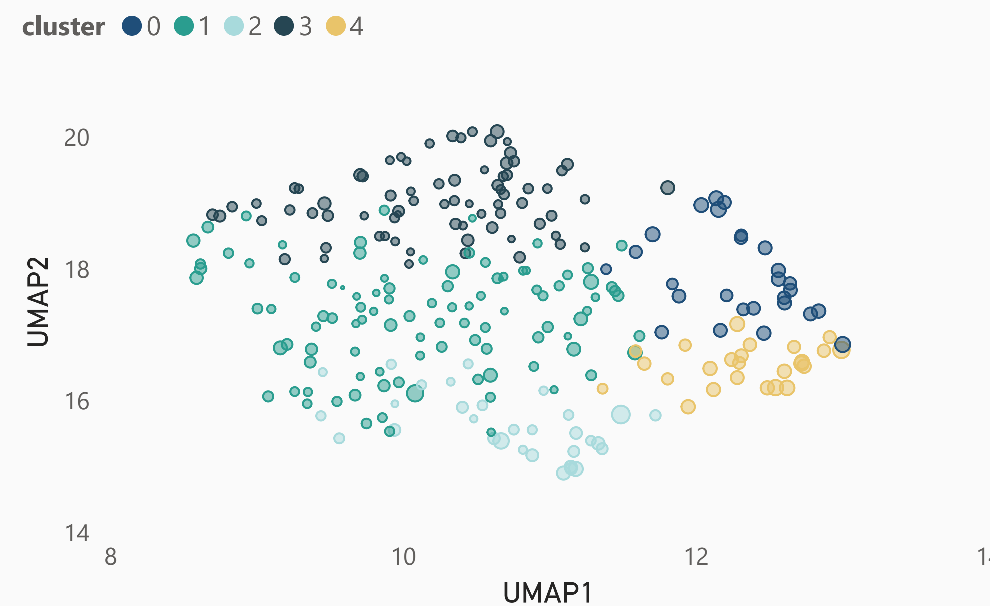
0 | 4

PCA Projection of Transcriptomic Signatures



Clusters show partial separation in the latent transcriptomic space.

UMAP Projection of Transcriptomic Signatures



Clusters & Compunds — Summary Overview

Cluster summaries reveal heterogeneity in size and quality metrics.

Filters

Compounds Name

Todas

Cell Lines

Todas

Dose

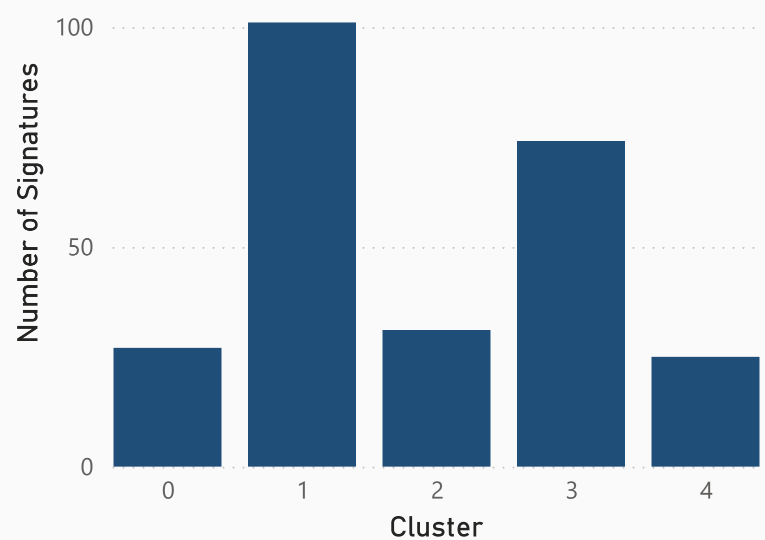
0,01 10,00

Clusters

0 4

cluster	n_signatures	n_compounds	n_cell_lines	mean_tas	mean_cc_q75	top_compounds	top_cell_lines
4	25	6	1	0,33	0,43	calcitriol (12); paricalcitol (4); calcipotriol (3)	PC3 (25)
0	27	7	1	0,32	0,44	calcitriol (7); seocalcitol (5); tacalcitol (5)	PC3 (27)
2	31	6	5	0,22	0,24	calcitriol (9); maxacalcitol (9); ercalcitriol (5)	A549 (11); HA1E (10); MCF7 (6)
3	74	7	5	0,16	0,23	calcitriol (31); maxacalcitol (11); ercalcitriol (8)	MCF7 (32); HA1E (20); A549 (18)
1	101	7	5	0,16	0,21	calcitriol (32); calcipotriol (15); tacalcitol (14)	MCF7 (32); A549 (27); HA1E (24)

Cluster Size (Number of Signatures)



Key Cluster Statistics

Largest cluster size	Highest mean TAS	Highest mean CC_Q75
101	0,33	0,44

Conclusions

Key Findings

- Transcriptomic signatures induced by vitamin D–related compounds organize into partially overlapping clusters, with non-linear embedding (UMAP) revealing clearer latent organization than linear projection (PCA).
- **Cluster size does not necessarily correlate with signal quality**, as measured by TAS and CC_Q75 metrics.
- Certain vitamin D analogs show **preferential association with specific clusters**, suggesting compound-driven transcriptomic structure.

Methodological Notes

- Clusters were derived from **unsupervised analysis** of transcriptomic signatures.
- Quality control metrics (TAS, CC_Q75) were used to assess **signature robustness and reproducibility**.

Limitations & Next Steps

- The limited number of cell lines constrains generalization across biological contexts.
- Future work may include **dose–response modeling**, **pathway-level enrichment**, or integration with additional omics layers.

Closing line

This dashboard provides a structured overview of transcriptomic responses to vitamin D–related compounds, serving as a foundation for downstream biological interpretation.